Chapter 9 Infinite Series

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March 19, 2026

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Polynomial approximations of elementary functions

- To find a polynomial function P that approximates another function f, begin by choosing a number c in the domain of f at which f and P have the same value. That is,

$$P(c) = f(c). \qquad \text{Graphs of } f \text{ and } P \text{ pass through } (c, f(c))$$

- The approximating polynomial is said to be **expanded about** c or **centered at** c.
- Geometrically, the requirement that $P(c) = f(c)$ means that the graph of P passes through the point $(c, f(c))$. Of course, there are many polynomials whose graphs pass through the point $(c, f(c))$.

- To find a polynomial whose graph resembles the graph of f near this point. One way to do this is to impose the additional requirement that the slope of the polynomial function be the same as the slope of the graph of f at the point $(c, f(c))$.

$$P'(c) = f'(c). \qquad \text{Graphs of } f \text{ and } P \text{ have the same at } (c, f(c))$$

- With these two requirements, you can obtain a simple linear approximation of f, as shown in Figure 1.

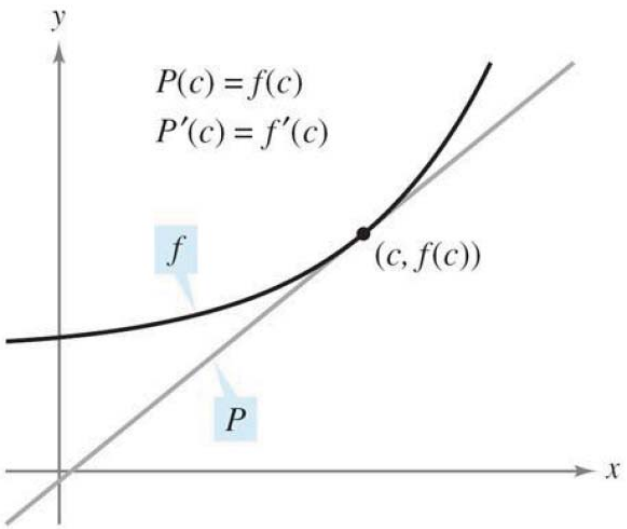


Figure 1: Near $(c, f(c))$, the graph of P can be used to approximate the graph of f.

Example 1 (First-degree polynomial approximation of $f(x) = e^x$)

For the function $f(x) = e^x$, find a first-degree polynomial function

$$P_1(x) = a_0 + a_1 x$$

whose value and slope agree with the value and slope of f at $x = 0$.

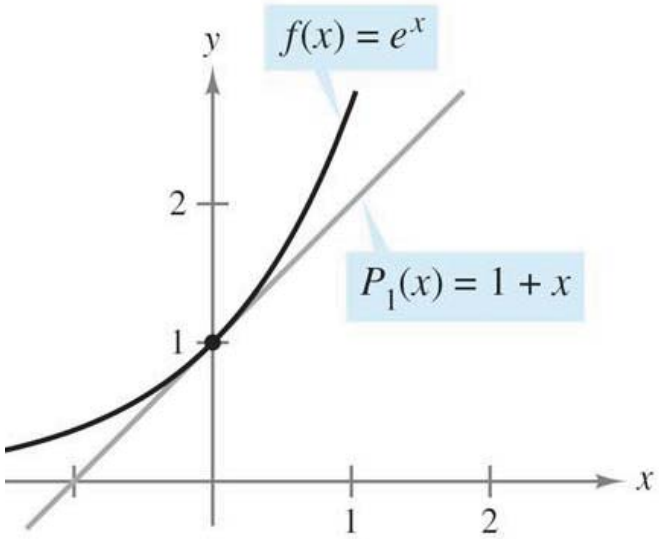


Figure 2: $P_1(x) = 1 + x$ is the first-degree polynomial approximation of $f(x) = e^x$.

Example 2 (Third-degree polynomial approximation of $f(x) = e^x$)

Construct a table comparing the values of the polynomial

$$P_3(x) = 1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3$$

with $f(x) = e^x$ for several values of x near 0.

x	-1.0	-0.2	-0.1	0	0.1	0.2	1.0
e^x	0.3679	0.81873	0.904837	1	1.105171	1.22140	2.7183
$P_3(x)$	0.3333	0.81867	0.904833	1	1.105167	1.22133	2.6667

Taylor and Maclaurin polynomials

- The polynomial approximation of $f(x) = e^x$ is expanded about $c = 0$. For expansions about an arbitrary value of c, it is convenient to write the polynomial in the form

$$P_n(x) = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots + a_n(x-c)^n.$$

- In this form, repeated differentiation produces

$$\begin{aligned}
P_n'(x) &= a_1 + 2a_2(x-c) + 3a_3(x-c)^2 + \cdots + na_n(x-c)^{n-1} \\
P_n''(x) &= 2a_2 + 2(3a_3)(x-c) + \cdots + n(n-1)a_n(x-c)^{n-2} \\
P_n'''(x) &= 2(3a_3) + \cdots + n(n-1)(n-2)a_n(x-c)^{n-3} \\
&\vdots \\
P_n^{(n)}(x) &= n(n-1)(n-2)\cdots(2)(1)a_n.
\end{aligned}$$

- Letting $x = c$, you then obtain

$$P_n(c) = a_0, \quad P_n'(c) = a_1, \quad P_n''(c) = 2a_2, \quad \ldots, \quad P_n^{(n)}(c) = n!a_n.$$

- Because the values of f and its first n derivatives must agree with the values of P_n and its first n derivatives at $x = c$, it follows that

$$f(c) = a_0, \quad f'(c) = a_1, \quad \frac{f''(c)}{2!} = a_2, \quad \ldots, \quad \frac{f^{(n)}(c)}{n!} = a_n.$$

Definition 9.1 (Taylor polynomial and Maclaurin polynomial)

If f has n derivatives at c, then the polynomial

$$P_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

is called the **nth Taylor polynomial for f at c**. If $c = 0$, then

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

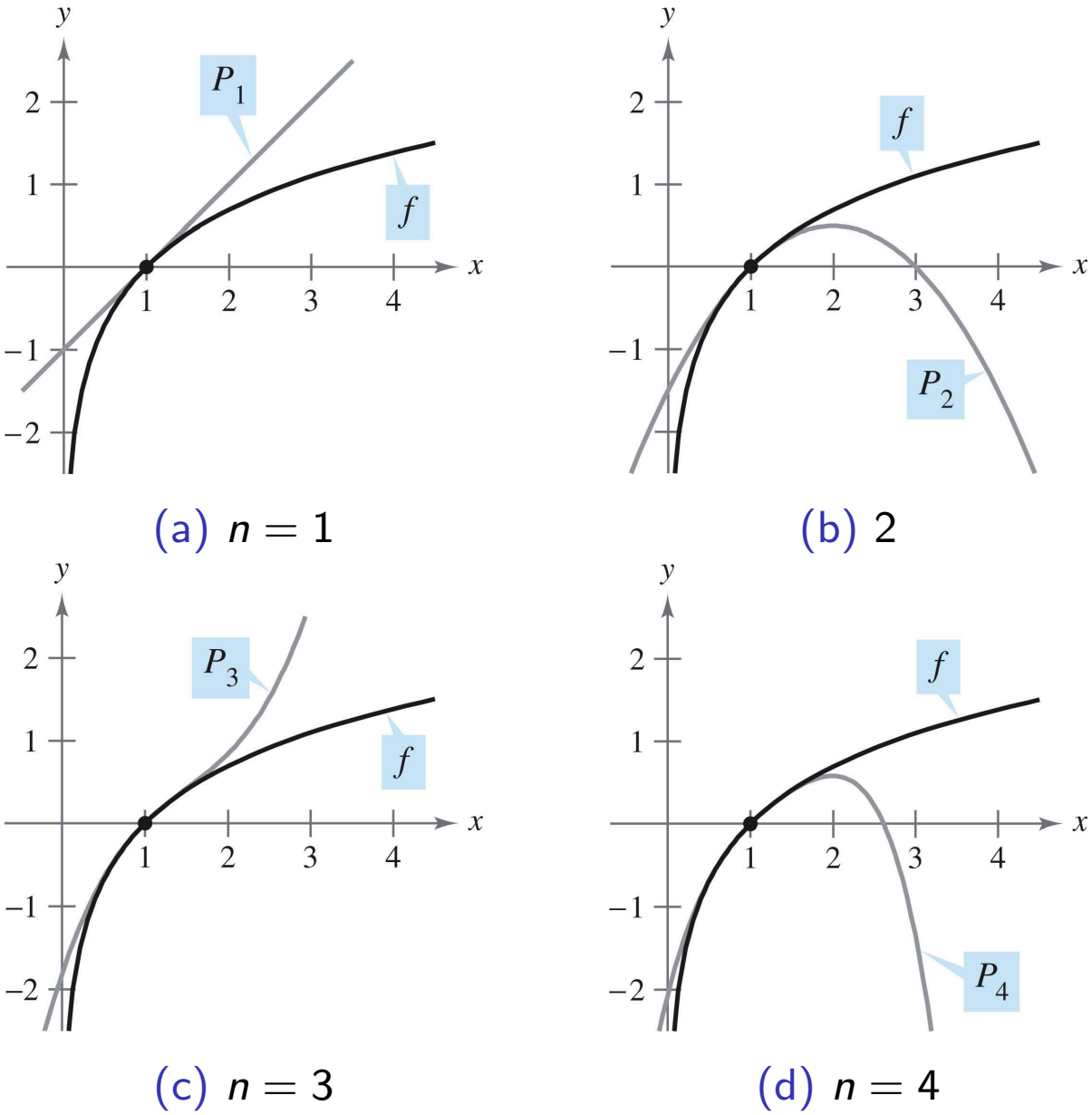
is also called the **nth Maclaurin polynomial for f**.

Example 3 (A Maclaurin polynomial for $f(x) = e^x$)

Find the nth Maclaurin polynomial for $f(x) = e^x$.

Example 4 (Finding Taylor polynomials for $\ln x$)

Find the Taylor polynomials P_0, P_1, P_2, P_3, and P_4, for $\ln x$ centered at $c = 1$.



(a) $n = 1$ (b) 2

(c) $n = 3$ (d) $n = 4$

Figure 3: As n increases, the graph of P_n, becomes a better and better approximation of the graph of $f(x) = \ln x$ near $x = 1$.

Example 5 (Finding Maclaurin polynomials for $\cos x$)

Find the Maclaurin polynomials P_0, P_2, P_4, and P_6 for $f(x) = \cos x$. Use $P_6(x)$ to approximate the value of $\cos(0.1)$.

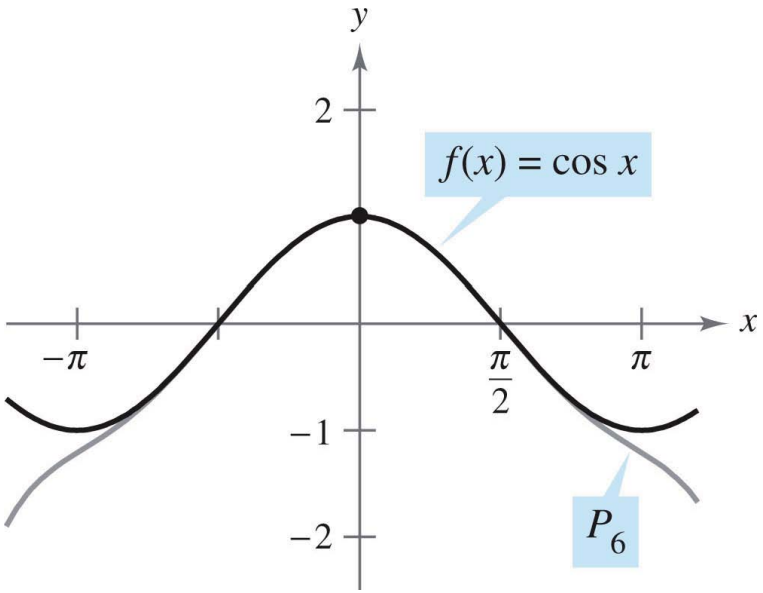


Figure 4: Near $(0, 1)$, the graph of P_6 can be used to approximate the graph of $f(x) = \cos x$.

Example 6 (Finding a Taylor polynomial for $\sin x$)

Find the third Taylor polynomial for $f(x) = \sin x$, expanded about $c = \pi/6$.

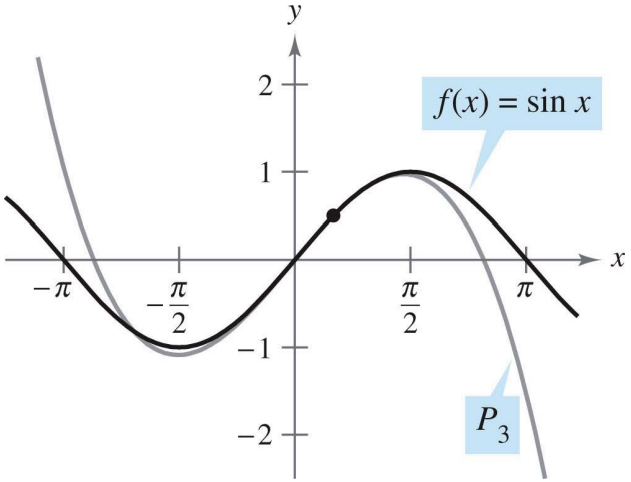


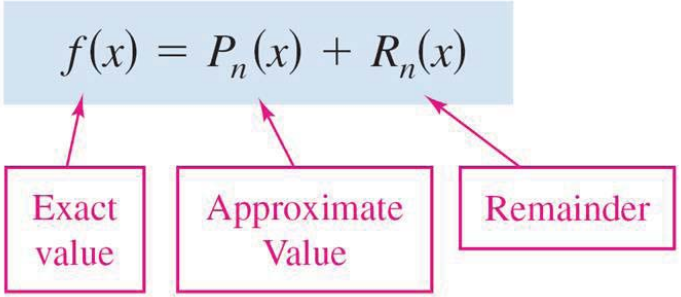
Figure 5: Near $(\pi/6, 1/2)$, the graph of P_3 can be used to approximate the graph of $f(x) = \sin x$.

Example 7 (Approximation using Maclaurin polynomials)

Use a fourth Maclaurin polynomial to approximate the value of $\ln(1.1)$.

Remainder of a Taylor polynomial

- An approximation technique is of little value without some idea of its accuracy. To measure the accuracy of approximating a function value $f(x)$ by the Taylor polynomial $P_n(x)$, you can use the concept of a remainder $R_n(x)$, defined as follows.



- So, $R_n(x) = f(x) - P_n(x)$. The absolute value of $R_n(x)$ is called the **error** associated with the approximation. That is,

$$\text{Error} = |R_n(x)| = |f(x) - P_n(x)|$$

- The next theorem gives a general procedure for estimating the remainder associated with a Taylor polynomial.

- This important theorem is called **Taylor's Theorem**, and the remainder given in the theorem is called the **Lagrange form of the remainder**.

Theorem 9.1 (Taylor's Theorem)

If a function f is differentiable through order $n+1$ in an interval I containing c, then, for each x in I, there exists z between x and c such that

$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + R_n(x)$$

where

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1}.$$

Example 8 (Determining the accuracy of an approximation)

The third Maclaurin polynomial for $\sin x$ is given by

$$P_3(x) = x - \frac{x^3}{3!}.$$

Use Taylor's Theorem to approximate $\sin(0.1)$ by $P_3(0.1)$ and determine the accuracy of the approximation.

- Using Taylor's Theorem, you have

$$\sin x = x - \frac{x^3}{3!} + R_3(x) = x - \frac{x^3}{3!} + \frac{f^{(4)}(z)}{4!}x^4$$

  where $0 < z < 0.1$.
- Therefore,

$$\sin(0.1) \approx 0.1 - \frac{(0.1)^3}{3!} \approx 0.1 - 0.000167 = 0.099833.$$

- Because $f^{(4)}(z) = \sin z$, it follows that the error $|R_3(0.1)|$ can be bounded as follows.

$$0 < R_3(0.1) = \frac{\sin z}{4!}(0.1)^4 < \frac{0.0001}{4!} \approx 0.000004.$$

- This implies that

$$\begin{aligned} 0.099833 < \sin(0.1) &= 0.099833 + R_3(x) < 0.099833 + 0.000004 \\ 0.099833 < \sin(0.1) &< 0.099837. \end{aligned}$$

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Sequences

- A **sequence** is a <u>function</u> whose domain is the set of positive integers. Although it is a function, it is common to represent sequences by subscript notation.
- For instance, in the sequence

$$\begin{array}{ccccccc} 1, & 2, & 3, & 4, & \ldots, & n, & \ldots \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ a_1, & a_2, & a_3, & a_4, & \ldots, & a_n, & \ldots \end{array} \quad \text{Sequence}$$

  1 is mapped onto a_1, 2 is mapped onto a_2, and so on.
- The numbers $a_1, a_2, \ldots, a_n, \ldots$ are the **terms** The number a_n is the n**th term** of the sequence, and the entire sequence is denoted by $\{a_n\}$.

Infinite series

- Infinite sequences can be used to represent infinite summations.
- Informally, if $\{a_n\}$ is an infinite sequence, then

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots + a_n + \cdots \qquad \text{Infinite series}$$

  is an **infinite series** (or simply a **series**).
- The numbers a_1, a_2, a_3, are the **terms** of the series.
- For some series, it is convenient to begin the index at $n = 0$ (or some other integer) and it is common to represent an infinite series as simply $\sum a_n$.

Power series

- An important function $f(x) = e^x$ can be represented <u>exactly</u> by an infinite series called a **power series**. For example, the power series representation for e^x is

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots .$$

- For each real number x, it can be shown that the infinite series on the right converges to the number e^x.

Definition 9.2 (Power series)

If x is a variable, then an infinite series of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n + \cdots$$

is called a **power series**. More generally, an infinite series of the form

$$\sum_{n=0}^{\infty} a_n (x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots + a_n(x-c)^n + \cdots$$

is called a **power series centered at** c, where c is a constant.

Example 1 (Power series)

a. The following power series is centered at 0.

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

b. The following power series is centered at -1.

$$\sum_{n=0}^{\infty} (-1)^n (x+1)^n = 1 - (x+1) + (x+1)^2 - (x+1)^3 + \cdots$$

c. The following power series is centered at 1.

$$\sum_{n=1}^{\infty} \frac{1}{n}(x-1)^n = (x-1) + \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + \cdots$$

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Differentiation and integration of power series

Theorem 9.2 (Properties of functions defined by power series)

If the function is given by
$f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots$
has a radius of convergence of $R > 0$*, then, on the interval*
$(c-R,\ c+R)$*,* f *is differentiable (and therefore continuous).*
Moreover, the derivative and antiderivative of f *are as follows.*

1. $f'(x) = \sum_{n=1}^{\infty} na_n(x-c)^{n-1} = a_1 + 2a_2(x-c) + 3a_3(x-c)^2 + \cdots$

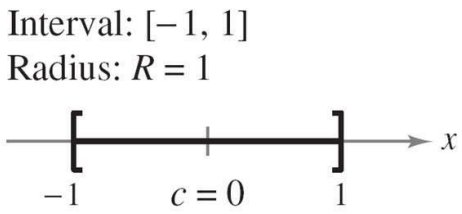
2.
$\int f(x)\,\mathrm{d}x = C + \sum_{n=0}^{\infty} a_n \frac{(x-c)^{n+1}}{n+1} = C + a_0(x-c) + a_1 \frac{(x-c)^2}{2} + a_2 \frac{(x-c)^3}{3} + \cdots$

Example 8 (Intervals of convergence for $f(x)$, $f'(x)$, and $\int f(x)\,\mathrm{d}x$)

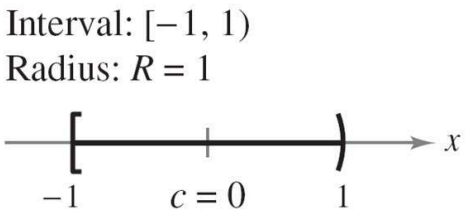
Consider the function given by

$$f(x) = \sum_{n=1}^{\infty} \frac{x^n}{n} = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots .$$

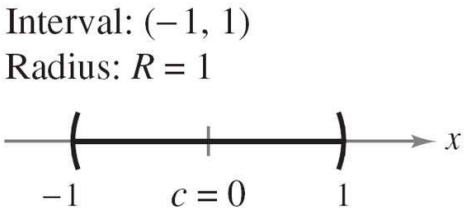
(Optional) Find the interval of convergence for each of the following.
a. $\int f(x)\,\mathrm{d}x$ **b.** $f(x)$ **c.** $f'(x)$



(a) Interval: $[-1, 1]$ and radius: $R = 1$.



(b) Interval: $[-1, 1)$ and radius: $R = 1$.



(c) Interval: $(-1, 1)$ and radius: $R = 1$.

Figure 6: Intervals of convergence for $f(x)$, $f'(x)$, and $\int f(x)\,\mathrm{d}x$.

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Geometric power series

- Consider the function given by $f(x) = 1/(1-x)$. The form of f closely resembles the sum of a geometric series

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}, \quad |r| < 1.$$

- In other words, if you let $a = 1$ and $r = x$, a power series representation for $1/(1-x)$, centered at 0, is

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots, \quad |x| < 1.$$

- Of course, this series represents $f(x) = 1/(1-x)$ only on the interval $(-1, 1)$, whereas f is defined for all $x \neq 1$, To represent f in another interval, you must develop a different series.

- For instance, to obtain the power series centered at -1, you could write

$$\frac{1}{1-x} = \frac{1}{2-(x+1)} = \frac{1/2}{1-[(x+1)/2]} = \frac{a}{1-r}$$

which implies that $a = 1/2$ and $r = (x+1)/2$.

- So, for $|x+1| < 2$, you have

$$\begin{aligned}\frac{1}{1-x} &= \sum_{n=0}^{\infty}\left(\frac{1}{2}\right)\left(\frac{x+1}{2}\right)^n \\ &= \frac{1}{2}\left[1 + \frac{(x+1)}{2} + \frac{(x+1)^2}{4} + \frac{(x+1)^3}{8} + \cdots\right], \quad |x+1| < 2\end{aligned}$$

which converges on the interval $(-3, 1)$.

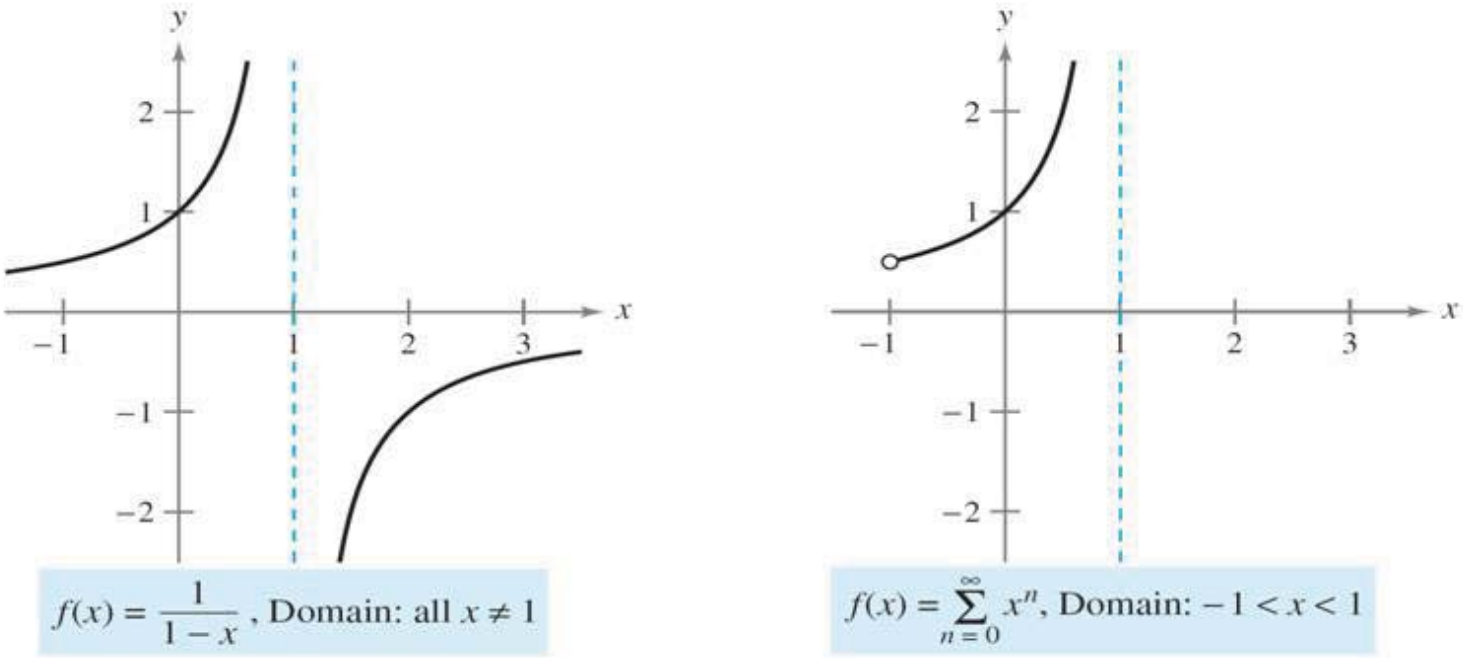

Figure 7: Definition of different ranges with function.

Example 1 (Finding a geometric power series centered at 0)

Find a power series for $f(x) = 4/(x+2)$ centered at 0.

Example 2 (Finding a geometric power series centered at 1)

Find a power series for $f(x) = 1/x$, centered at 1.

Operations with power series

Operations with power series Let $f(x) = \sum a_n x^n$ and $g(x) = \sum b_n x^n$.

1. $f(kx) = \sum_{n=0}^{\infty} a_n k^n x^n$

2. $f(x^N) = \sum_{n=0}^{\infty} a_n x^{nN}$

3. $f(x) \pm g(x) = \sum_{n=0}^{\infty} (a_n \pm b_n) x^n$

Example 3 (Adding two power series)

Find a power series, centered at 0, for $f(x) = (3x - 1)/(x^2 - 1)$.

Example 4 (Finding a power series by integration)

Find a power series for $f(x) = \ln x$, centered at 1.

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Taylor series and Maclaurin series

- The development of power series to represent functions is credited to the combined work of many seventeenth and eighteenth-century mathematicians.
- However, the two names that are most commonly associated with power series are Brook Taylor and Colin Maclaurin.

Theorem 9.3 (The form of a convergent power series)

If f is represented by a power series $f(x) = \sum a_n(x-c)^n$ for all x in an open interval I containing c, then $a_n = f^{(n)}(c)/n!$ and

$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \cdots .$$

- The coefficients of the power series in Theorem 9.3 are precisely the coefficients of the Taylor polynomials for $f(x)$ at c. For this reason, the series is called the Taylor series for $f(x)$ at c.

Definition 9.3 (Taylor and Maclaurin series)

If a function f has derivatives of all orders at $x = c$, then the series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n = f(c) + f'(c)(x-c) + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \cdots$$

is called the **Taylor series for** $f(x)$ **at** c. Moreover, if $c = 0$, then the series is the **Maclaurin series for** f.

Example 1 (Forming a power series)

Use the function $f(x) = \sin x$ to form the Maclaurin series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \frac{f^{(3)}(0)}{3!} x^3 + \frac{f^{(4)}(0)}{4!} x^4 + \cdots$$

- You cannot conclude that the power series converges to $\sin x$ for all x. You can simply conclude that the power series converges to some function, but you are not sure what function it is. This is a subtle, but important, point in dealing with Taylor or Maclaurin series.
- To persuade yourself that the series

$$f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \cdots$$

  might converge to a function other than f, remember that the derivatives are being evaluated at a single point.
- It can easily happen that another function will agree with the values of $f^{(n)}(x)$ when $x = c$ and disagree at other x-values.

- If you formed the power series for the function shown in Figure 8, you would obtain the same series as in Example 1. You know that the series converges for all x, and yet it obviously cannot converge to both $f(x)$ and $\sin x$ for all x.

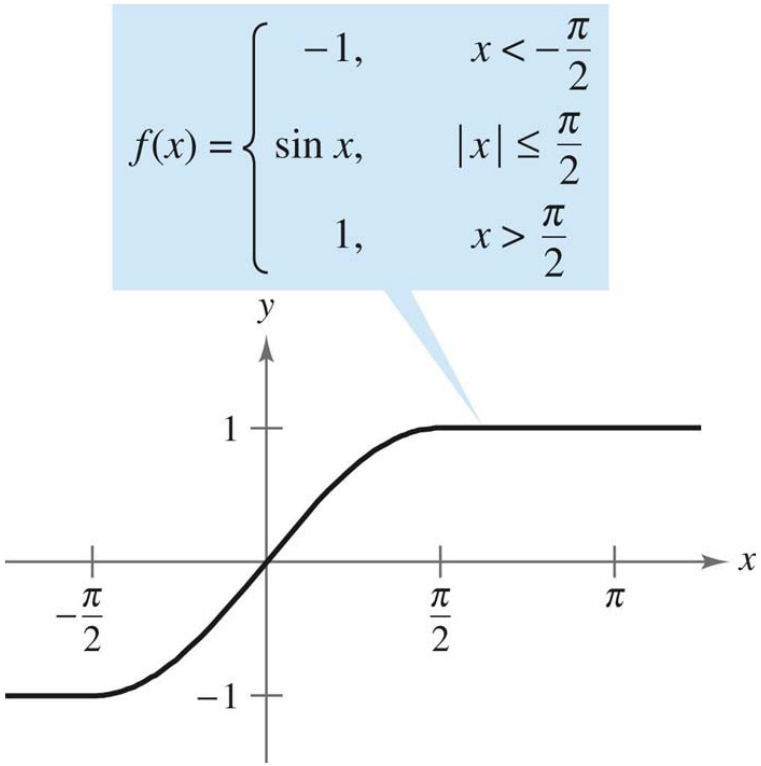


Figure 8: $f(x) \neq \sin x$ for all x but both have the same Taylor series.

- Let f have derivatives of all orders in an open interval I centered at c.
- The Taylor series for f may fail to converge for some x in I. Or, even if it is convergent, it may fail to have $f(x)$ as its sum.
- Nevertheless, Theorem 9.1 tells us that for each n,

$$f(x) = f(c)+f'(c)(x-c)+\frac{f''(c)}{2!}(x-c)^2+\cdots+\frac{f^{(n)}(c)}{n!}(x-c)^n+R_n(x)$$

where

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1}.$$

Theorem 9.4 (Convergence of Taylor series)

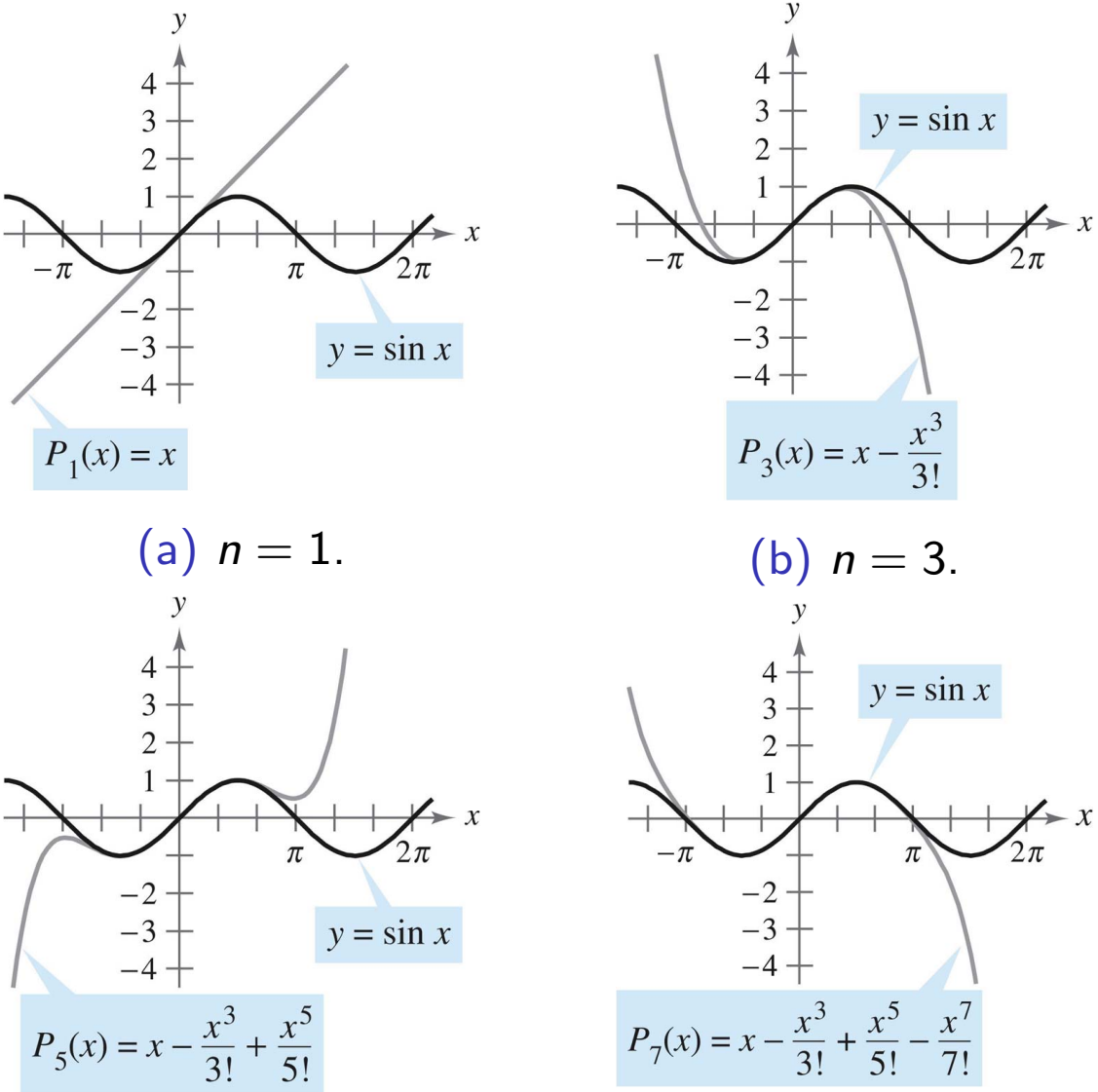
If $\lim_{n\to\infty} R_n = 0$ *for all* x *in the interval* I*, then the Taylor series for* f *converges and equals* $f(x)$*,*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n.$$

Example 2 (A convergent Maclaurin series)

Show that the Maclaurin series for $f(x) = \sin x$ converges to $\sin x$ for all x.

- Figure 9 visually illustrates the convergence of the Maclaurin series for $\sin x$ by comparing the graphs of the Maclaurin polynomials $P_1(x)$, $P_3(x)$, $P_5(x)$, and $P_7(x)$ with the graph of the sine function.



(a) $n = 1$. (b) $n = 3$.

(c) $n = 5$. (d) $n = 7$.

Figure 9: As n increases, the graph of P_n more closely resembles the sine function.

Guidelines for finding a Taylor series

1. Differentiate $f(x)$ several times and evaluate each derivative at c.

$$f(c), \quad f'(c), \quad f''(c), \quad f'''(c), \quad \ldots, \quad f^{(n)}(c), \quad \ldots$$

   Try to recognize a pattern in these numbers.

2. Use the sequence developed in the first step to form the Taylor coefficients $a_n = f^{(n)}(c)/n!$, and determine the interval of convergence for the resulting power series

$$f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \cdots .$$

3. Within this interval of convergence, determine whether or not the series converges to $f(x)$.

Example 3 (Maclaurin series for a composite function)

Find the Maclaurin series for $f(x) = \sin x^2$.

Binomial series

Example 4 (Binomial series)

Find the Maclaurin series for $f(x) = (1+x)^k$. Assume that k is not a positive integer.

Example 5 (Finding a binomial series)

Find the Maclaurin series for $f(x) = \sqrt[3]{1+x}$.

Deriving Taylor series from a basic list

Function	Interval of convergence
$\frac{1}{x} = 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4 - \cdots + (-1)^n (x-1)^n + \cdots$	$0 < x < 2$
$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + \cdots + (-1)^n x^n + \cdots$	$-1 < x < 1$
$\ln x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \cdots + \frac{(-1)^{(n-1)}(x-1)^n}{n} + \cdots$	$0 < x \leq 2$
$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots + \frac{x^n}{n!} + \cdots$	$-\infty < x < \infty$
$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \cdots$	$-\infty < x < \infty$
$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots + \frac{(-1)^n x^{2n}}{(2n)!} + \cdots$	$-\infty < x < \infty$
$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots + \frac{(-1)^n x^{2n+1}}{2n+1} + \cdots$	$-1 \leq x \leq 1$
$\arcsin x = x + \frac{x^3}{2 \cdot 3} + \frac{1 \cdot 3 x^5}{2 \cdot 4 \cdot 5} + \frac{1 \cdot 3 \cdot 5 x^7}{2 \cdot 4 \cdot 6 \cdot 7} + \cdots + \frac{(2n)! x^{2n+1}}{(2^n n!)^2 (2n+1)} + \cdots$	$-1 \leq x \leq 1$
$(1+x)^k = 1 + kx + \frac{k(k-1)x^2}{2!} + \frac{k(k-1)(k-2)x^3}{3!} + \frac{k(k-1)(k-2)(k-3)x^4}{4!} + \cdots$	$-1 < x < 1$

The conevrgence at $x = \pm 1$ depends on the value of k.

Euler's Formula

$$e^{ix} = \cos x + i \sin x = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!}$$
$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} + i \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

Example 6 (Deriving a power series from a basic list)

Find the power series for $f(x) = \cos\sqrt{x}$.

Example 7 (Multiplication and division of power series)

Find the first three nonzero terms in each Maclaurin series $e^x \arctan x$.

Example 8 (Division of Power Series)

Find the first three nonzero terms in each Maclaurin series $\tan x$.

Example 9 (A power series for $\sin^2 x$)

Find the power series for $f(x) = \sin^2 x$.

Example 10 (Power series approximation of a definite integral)

Use a power series to approximate

$$\int_0^1 e^{-x^2}\,\mathrm{d}x$$

with an error of less than 0.01.